A+ Computer Science
M/C Written Test

General Directions:

1) DO NOT OPEN EXAM UNTIL TOLD TO DO SO.

2) **NO CALCULATORS of any kind may be used.**

3) You have 45 minutes to complete this contest. If you are in the process of actually writing an answer when the signal to stop is given, you may finish writing that answer.

4) Papers may not be turned in until forty-five minutes have elapsed. If you finish the test before the end of the allotted time, remain at your seat and retain your paper until told to do otherwise. You may use this time to check your answers.

5) All answers must be written on the answer sheet/Scantron card provided. Indicate your answers in the appropriate blanks provided on the answer sheet or on the Scantron card. Clean erasures are necessary for accurate Scantron grading.

6) You may place as many notations as you desire anywhere on the test paper except on the answer sheet or Scantron card which is reserved for answers only.

7) You may use additional scratch paper provided by the contest director.

8) All questions have ONE and only ONE correct (BEST) answer. There is a penalty for all incorrect answers. **All provided code segments are intended to be syntactically correct, unless otherwise stated (i.e. `error` is an answer choice). Ignore any typographical errors and assume any undefined variables are defined as used.**

9) A reference to commonly used Java classes is provided with the test and you may use this reference during the contest. You may detach the reference sheets from the test booklet but DO NOT DO SO UNTIL THE CONTEST BEGINS.

10) Assume that any necessary import statements for Standard Java 23 Packages and classes (e.g. .lang, .util, System, Math, Double, etc.) are included in any programs or code segments that refer to methods from these classes and/or packages.

Scoring:

1) All questions will receive 6 points if answered correctly; no points will be given or subtracted if unanswered; 2 points will be deducted for each incorrect answer.

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Standard Classes and Interfaces — Supplemental Reference

class java.lang.Object
- boolean equals(Object other)
- String toString()
- int hashCode()

interface java.lang.Comparable<T>
- int compareTo(T other)
  Return value < 0 if this is less than other.
  Return value = 0 if this is equal to other.
  Return value > 0 if this is greater than other.

class java.lang.Integer implements Comparable<Integer>
- Integer(int value)
- int intValue()
- boolean equals(Object obj)
- String toString()
- int compareTo(Integer anotherInteger)
- static int parseInt(String s)

class java.lang.Double implements Comparable<Double>
- Double(double value)
- double doubleValue()
- boolean equals(Object obj)
- String toString()
- int compareTo(Double anotherDouble)
- static double parseDouble(String s)

class java.lang.String implements Comparable<String>
- int compareTo(String anotherString)
- boolean equals(Object obj)
- int length()
- String substring(int begin, int end)
  Returns the substring starting at index begin and ending at index (end - 1).
- String substring(int begin)
  Returns substring(from, length()).
- int indexOf(String str)
  Returns the index within this string of the first occurrence of str. Returns -1 if str is not found.
- int indexOf(String str, int fromIndex)
  Returns the index within this string of the first occurrence of str, starting the search at the specified index.. Returns -1 if str is not found.
- charAt(int index)
- int indexOf(int ch)
- int indexOf(int ch, int fromIndex)
- String toLowerCase()
- String toUpperCase()
- String[] split(String regex)
- boolean matches(String regex)

class java.lang.Character
- static boolean isDigit(char ch)
- static boolean isLetter(char ch)
- static boolean isLetterOrDigit(char ch)
- static boolean isLowerCase(char ch)
- static boolean isUpperCase(char ch)
- static char toUpperCase(char ch)
- static char toLowerCase(char ch)

class java.lang.Math
- static int abs(int a)
- static double abs(double a)
- static double pow(double base, double exponent)
- static double sqrt(double a)
- static double ceil(double a)
- static double floor(double a)
- static double min(double a, double b)
- static double max(double a, double b)
- static int min(int a, in b)
- static int max(int a, int b)
- static long round(double a)
- static double random()
  Returns a double value with a positive sign, greater than or equal to 0.0 and less than 1.0.

interface java.util.List<E>
- boolean add(E e)
- int size()
- Iterator<E> iterator()
- ListIterator<E> listIterator()
- E get(int index)
- E set(int index, E e)
  Replaces the element at index with the object e.
- void add(int index, E e)
  Inserts the object e at position index, sliding elements at position index and higher to the right (adds 1 to their indices) and adjusts size.
- E remove(int index)
  Removes element from position index, sliding elements at position (index + 1) and higher to the left (subtracts 1 from their indices) and adjusts size.

class java.util.ArrayList<E> implements List<E>

class java.util.LinkedList<E> implements List<E>, Queue<E>

Methods in addition to the List methods:
- void addFirst(E e)
- void addLast(E e)
- E getFirst()
- E getLast()
- E removeFirst()
- E removeLast()

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class java.util.Stack<E>
- boolean isEmpty()
- E peek()
- E pop()
- E push(E item)

interface java.util.Queue<E>
- boolean add(E e)
- boolean isEmpty()
- E peek()
- E remove()

class java.util.PriorityQueue<E>
- boolean add(E e)
- boolean isEmpty()
- E peek()
- E remove()

interface java.util.Set<E>
- boolean add(E e)
- boolean contains(Object obj)
- boolean remove(Object obj)
- int size()
- Iterator<E> iterator()
- boolean addAll(Collection<? extends E> c)
- boolean removeAll(Collection<?> c)
- boolean retainAll(Collection<?> c)

class java.util.HashSet<E> implements Set<E>

class java.util.TreeSet<E> implements Set<E>

interface java.util.Map<K,V>
- Object put(K key, V value)
- V get(Object key)
- boolean containsKey(Object key)
- int size()
- Set<K> keySet()
- Set<Map.Entry<K, V>> entrySet()

class java.util.HashMap<K,V> implements Map<K,V>

class java.util.TreeMap<K,V> implements Map<K,V>

interface java.util.Map.Entry<K,V>
- K getKey()
- V getValue()
- V setValue(V value)

interface java.util.Iterator<E>
- boolean hasNext()
- E next()
- void remove()

interface java.util.ListIterator<E> extends java.util.Iterator<E>

Methods in addition to the `Iterator` methods:
- void add(E e)
- void set(E e)

class java.lang.Exception
- Exception()
- Exception(String message)

class java.util.Scanner
- Scanner(InputStream source)
- boolean hasNext()
- boolean hasNextInt()
- boolean hasNextDouble()
- String next()
- int nextInt()
- double nextDouble()
- String nextLine()
- Scanner useDelimiter(String pattern)

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Note: Correct responses are based on **Java JDK 23** from Oracle, Inc. All provided code segments are intended to be syntactically correct, unless otherwise stated (e.g., "error" is an answer choice) and any necessary Java JDK 23 Standard Packages have been imported. Ignore any typographical errors and assume any undefined variables are defined as used. **For all output statements, assume that the System class has been statically imported using: import static java.lang.System.***

QUESTION 1

Which of these values is the greatest?

A. 1303_7 B. 4210_5 C. 134_{13} D. 253_{11} E. $22A_{16}$

QUESTION 2

What is output by the code to the right?

A. 9 B. 12 C. 22 D. 8 E. 8.14

```
out.println(52 - 20 * 2);
```

QUESTION 3

What is output by the code to the right?

A. WhaUp%

B. WhatUpUp\%

C. WhaUp\\%

D. Wh aUp\%

E. There is no output due to a runtime error.

```
Out.printf("%.3s%.2s\\%","What","Up");
```

QUESTION 4

What is output by the code to the right?

A. true B. false C. null D. Mountain

E. There is no output due to a compile error.

```
String a = "Mountain";
String b = "Mountain";
out.println( a==b );
```

QUESTION 5

How many unique combinations of x,y, and z will result in a being true?

A. 6

B. 4

C. 1

D. 5

E. 3

```
boolean x,y,z;
boolean a = !( x && y || !(z ^ y));
```

QUESTION 6

What is output by the code to the right?

A. 128.0 B. 256 C. 129 D. 129.0

E. There is no output due to a compile error.

```
short tall = 101;
double wide = 27;
out.println(Math.ceil(tall+wide) );
```

QUESTION 7

What is output by the code to the right?

A. 53 B. 100 C. 49 D. 47

E. There is no output due to a compile error.

```
int x,y,z;
x=7; y=10; z=4;
out.println(x+y*z);
```

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QUESTION 8

What is output by the code to the right?

A. `OneDefaultFiveFour`

B. `DefaultFiveFour`

C. `DefaultFourFive`

D. `FiveFour`

E. `Default`

```
int num = 2; String a ="";
switch (num+1) {
    case 1:
        a+="Two"; num+=1;
    case 2:
        a+="One"; num-=2;
    default:
        out.print("Default");
    case 4:
        a+="Five"; num+=4;
    case 5:
        a+="Four"; num-=5; break;
}
out.println(a);
```

QUESTION 9

How many asterisks are output by the code to the right?

A. `6` B. `4` C. `5` D. `9` E. `0`

```
int z = 0;
for (int i = 0; i < 20; i+=4)
   z++;
out.print(z);
```

QUESTION 10

What is the smallest value of `x` will result in the greatest possible ending value of `y`?

A. `77`

B. `33`

C. `8`

D. `-1`

E. `null`

```
int[] x = {3,5,8,33,77,-33,11,456};
out.print(x[7/2]);
```

QUESTION 11

What is the output by the code to the right?

A. `How`

B. `you`

C. `doin?`

D. There is no output due to a compile error.

E. There is no output due to a runtime exception.

```
String joey = "How you doin?";
Scanner scan = new Scanner(joey);
scan.next();
scan.next();
out.println(scan.next());
```

QUESTION 12

What could replace **<1*>** in order for the loop to run infinitely?

A. `b > -100`

B. `c % 3 == 1`

C. `c > 0`

D. `b != 0`

E. None of the above

```
int a = 5;
int b = 3;
int c = 10;

while ( <*1> ) {
    a = a + b;
    b = b - 1;
    c = c + 2;
}
```

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QUESTION 13

What is order of precedence for the operations on the right from highest precedence to lowest precedence?

A. I, II, III B. I, III, II

C. II, III, I D. III, I, II

E. III, II, I

I. <<(Bitwise left shift)
II. >= (additive)
III. ! (Unary logical negation)

QUESTION 14

What is the output by the code to the right?

A. 8 B. 64 C. 32 D. 16 E. 4

```
out.println(Integer.BYTES*Short.BYTES);
```

QUESTION 15

What is the output by the code to the right?

A. [1, 2, 5, 3, true]

B. [1, 4, 2, 0, 3, 5]

C. [1, 4, 3, 5]

D. [1, 4, 0, 3, true]

E. There is no output due to a runtime exception.

```
ArrayList<Object> a;
a = new ArrayList<>();
a.add(1); a.add(4); a.add(2); a.add(3);
a.add(0); a.add(5);
a.add(a.remove(a.indexOf(3)));
a.add(a.remove(a.get(1)));
a.remove(a.remove(2));
out.println(a);
```

QUESTION 16

What is the output by the code to the right?

A. 25 B. 17 C. 63 D. 9 E. 43

```
out.println(43 & 29 | 17);
```

QUESTION 17

Which of these values is the greatest and could be generated by the expression on the right?

A. 52.1 B. 46.9 C. 51 D. 25.0 E. 24.9

```
out.println(Math.random()*46-Math.random()*27+6);
```

QUESTION 18

Which of the following lines takes longer to execute?

A. Line 1

B. Line 2

C. Both lines take the same amount of time

D. They can take longer than the other, it depends

E. It is impossible to know which line takes longer

```
ArrayList<Integer> a;
a = new ArrayList<>();
a.add(6);
a.add(4);
a.add(3);
Collections.sort(a); //Line 1
Collections.sort(a); //Line 2
```

QUESTION 19

What is the output by the code to the right?

A. 247Lots247 B. 49Lots247 C. 49Lots49

D. There is no output due to a compile error.

E. There is no output due to a runtime exception.

```
String a = "Lots";
int g = 47;
out.println(2+g+a+2+g);
```

QUESTION 20

How many unique variable combinations result in the following expression being false?

A. 9 B. 10 C. 11 D. 12 E. 13

$$(A+B)(\overline{C+D})$$

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QUESTION 21

What is the output of the function call `uptown(12, 7)`?

A. `8`

B. `9`

C. `7`

D. `5`

E. There is no output due to a stack overflow error.

QUESTION 22

What is the output of the function call `funk(19, 12)`?

A. `14`

B. `12`

C. `11`

D. `13`

E. There is no output due to a stack overflow error.

```
// Use the code below to answer
// questions 21-22.

public int uptown(int a, int b){
    if(a>b){
        return funk(b-1,a-2)+1;
    }
    if(b>a){
        return funk(a-1,b-2)+2;
    }
    return -1;
}
public int funk(int x, int y){
    if(x>y){
        return uptown(y,x-2)+1;
    }
    if(y>x){
        return uptown(x-2,y-3)+3;
    }
    return -2;
}
```

QUESTION 23

What should replace **<code 1>** in the code to the right in order for `employee` to be a subclass of `person` and `boss` to be a subclass of `employee`?

A. `defines`

B. `extends`

C. `implements`

D. `inherits`

E. Nothing, this space can be left blank.

QUESTION 24

What should replace **<code 2>** in the code to the right in order for `boss` constructor to compile?

A. `super(n,w);`

B. `super(n,s);`

C.
```
super(n,w);
swagLevel=s;
```

D.
```
super(n,s);
swagLevel=w;
```

E. All of the above

```
// Use the code below to answer
// questions 23-28.

class person{
    String name;
    public person(String n){
        name=n;
    }
}
class employee <code 1> person{
    double wage;
    public employee(String n, double w){
        super(n);
        wage=w;
    }
    public void work(){
        out.println("Working");
    }
    public void work(int x){
        out.println("Working for "+x+" hours");
    }
}
class boss <code 1> employee{
    int swagLevel;
    public boss (String n, double w, int s){
        <code 2>
    }
    public void work(){
        int x = 8-swagLevel;
        out.println("Working for "+x+" hours");
    }
}
```

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QUESTION 25

Assuming the blanks above have been properly filled and any errors fixed, the second work function in the `employee` class demonstrates what principle?

A. Encapsulation

B. Abstraction

C. Inheritance

D. Polymorphism

E. Overriding

QUESTION 26

Assuming the blanks above have been properly filled and any errors fixed, the work function in the `boss` class demonstrates what principle?

A. Encapsulation

B. Abstraction

C. Inheritance

D. Polymorphism

E. Overriding

```
// Use the code below to answer
// questions 23-28.
class person{
    String name;
    public person(String n){
        name=n;
    }
}
class employee <code 1> person{
    double wage;
    public employee(String n, double w){
        super(n);
        wage=w;
    }
    public void work(){
        out.println("Working");
    }
    public void work(int x){
        out.println("Working for "+x+" hours");
    }
}
class boss <code 1> employee{
    int swagLevel;
    public boss (String n, double w, int s){
        <code 2>
    }
    public void work(){
        int x = 8-swagLevel;
        out.println("Working for "+x+" hours");
    }
}
```

QUESTION 27

Assuming the blanks above have been properly filled and any errors fixed, which line causes a compilation error?

A. `Line 1`

B. `Line 2`

C. `Line 3`

D. `Line 4`

E. None of the above

QUESTION 28

Assuming the blanks above have been properly filled and any errors fixed, what is printed as a result of `Line 5`?

A. `Working`

B. `Working for 31 hours`

C. `Working for 5 hours`

D. `Working for 29 hours`

E. There is no output due to a runtime exception.

```
//CLIENT CODE
// Use to answer questions 27-28

person mike = new person("mike"); //Line 1
person sally = new employee("sally",10.01); //Line 2
boss jimmy = new employee("jimmy",31.0); //Line 3
boss lauren = new boss("lauren", 31.0, 3); //Line 4
lauren.work(); //Line 5
```

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QUESTION 29

What is output by the code to the right?

A. `Infinity` B.`MAX VALUE` C. `3.4028235E38` D. `1.4E-45`

E. There is no output due to a runtime error.

```
float x = Float.MAX_VALUE;
x+=Float.MAX_VALUE;
out.println(x);
```

QUESTION 30

What is output by the code to the right?

A. `[3, 7, 8, 9, 10, 13]`

B. `[7, 13, 8, 9, 10, 3]`

C. `[3, 9, 7, 13, 10, 8]`

D. `[13, 8, 9, 10, 5, 3]`

E. Output cannot be determined until runtime.

```
PriorityQueue<Integer> pq;
pq = new PriorityQueue<>();
pq.add(7);
pq.add(13);
pq.add(8);
pq.add(9);
pq.add(10);
pq.add(5);
pq.remove();
pq.add(3);
out.println(pq);
```

QUESTION 31

What is output by the code to the right?

A. `2`

B. `25`

C. `6`

D. `13`

E. There is no output due to a runtime error.

```
int[][] mat = new int[4][5];
for (int i = 0; i < 4; i++) {
    for (int j = 0; j < 5; j++) {
        mat[i][j]=i*j;
        if(i>0) mat[i][j]-=mat[i-1][j];
        if(j>0) mat[i][j]+=mat[i][j-1];
    }
}
out.println(mat[3][3]);
```

QUESTION 32

What is output by `Line 1` in the code to the right?

A. `[4, 2, 2, 5]`

B. `[4, 2, 9, 2, 5]`

C. `[4, 3, 8, 9, 3, 2]`

D. `[4, 3, 3, 2]`

E. There is no output due to a runtime exception.

QUESTION 33

What is output by `Line 2` in the code to the right?

A. `[6, 8, 9, 3, 2]`

B. `[6, 8, 9, 3]`

C. `[2, 6, 8, 9, 3, 2]`

D. `[8, 9, 3]`

E. There is no output due to a runtime exception.

QUESTION 34

What is output by `Line 3` in the code to the right?

A. `[2, 5]`

B. `[5 ,2]`

C. `[8, 9, 3]`

D. `[2, 8, 9]`

E. There is no output due to a runtime exception.

```
//Use for questions 32-34
int[] arr = {4, 2, 6, 8, 9, 3, 2, 5};
ArrayList<Integer> a = new ArrayList<>();
Stack<Integer> s = new Stack<>();
Queue<Integer> q = new LinkedList<>();
for(int x: arr){
    a.add(x);
    s.push(x);
    q.add(x);
}
a.remove(s.pop());
s.remove(q.poll());
q.remove(a.set(2,3));
a.remove(s.getFirst());
s.remove(q.poll());
q.remove(a.remove(2));
a.remove(q.poll());
s.remove(a.get(q.peek()));
q.poll();
out.println(a); //Line 1
out.println(s); //Line 2
out.println(q); //Line 3
```

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QUESTION 35

What is output by the code to the right?

A. 15 B. 9 C. 12 D. 16

E. There is no output due to a compile error.

```
out.println(45 % 7 & 13 + 7 | 12);
```

QUESTION 36

What is the worst time complexity for a Quicksort on an array of size n?

A. O(1) B.O(n) C.O(log(n)) D. O(nlog(n)) E. O(n^2)

QUESTION 37

Which of the following Java boolean expressions is equivalent to expression on the right?

$$A+BC+\overline{D}$$

A. A || B && C || !D

B. A && B || C && !D

C. A && (B || C) ^ D

D. (A || B)&& C) ^ D

E. A || B || C || D

QUESTION 38

On the graph to the right, what is the cost of the most expensive path from node 8 to node 6 that does not cross the same node twice?

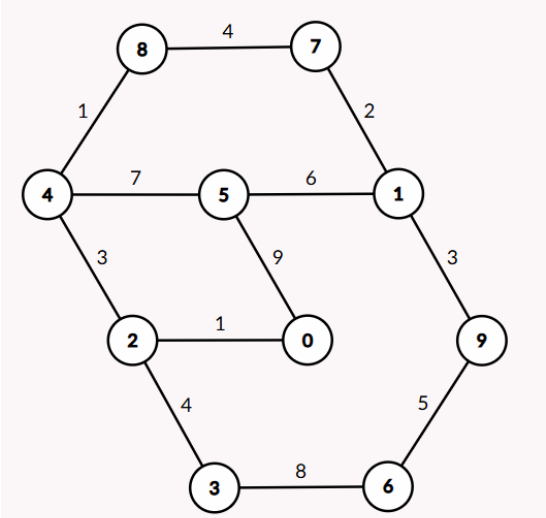
A. 20

B. 28

C. 30

D. 32

E. 34



QUESTION 39

What is the value of the prefix expression to the below? Assume integer division.

```
+ - * 5 4 / 11 3 7
```

QUESTION 40

What is the best case complexity for a search of a BST with a height of 20? (Big O notation)

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